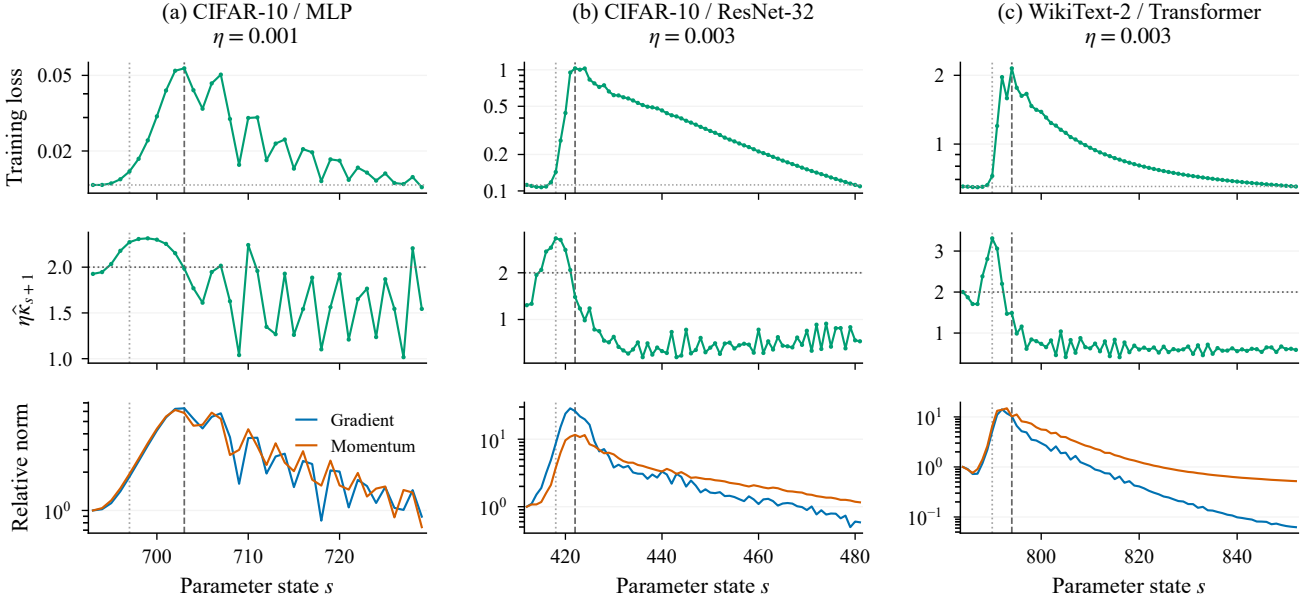

(a) CIFAR-10 / MLP
$\eta = 0.001$
(b) CIFAR-10 / ResNet-32
$\eta = 0.003$
(c) WikiText-2 / Transformer
$\eta = 0.003$
Training loss
$\eta\hat{\kappa}_{s+1}$
Relative norm
Gradient
Momentum
Parameter state s
Parameter state s
Parameter state s